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Electrolyzing synthesis of boron-doped graphene quantum dots for fluorescence determination of Fe^{3+} ions in water samples

Li Chen¹, Chuanli Wu¹, Pan Du, Xiaowei Feng, Ping Wu*, Chenxin Cai*

Jiangsu Key Laboratory of New Power Batteries, Jiangsu Collaborative Innovation Center of Biomedical Functional Materials, National and Local Joint Engineering Research Center of Biomedical Functional Materials, College of Chemistry and Materials Science, Nanjing Normal University, Nanjing 210023, P.R. China.

wuping@njnu.edu.cn (P. Wu)

cxcai@njnu.edu.cn (C. Cai)

*Corresponding authors. Tel.: +86 25 85891780; fax: +86 25 85891767.

Abstract

This work reports a facile electrolyzing method to synthesize boron-doped graphene quantum dots (BGQDs) and uses the BGQDs as a fluorescent probe to determine Fe^{3+} ion levels in water samples. The BGQDs were produced by oxidizing graphite in an aqueous borax solution at pH 7; then, the borate solution was filtered with BGQDs, and the borate was dialyzed from the filtrate, leaving a solution of BGQDs in water. The amount of the B in the BGQDs can be adjusted by changing the concentration of borax used for the electrolyte. The excitation wavelength- and B amount-dependent fluorescence characteristics of BGQDs were studied. The fluorescence intensity of the BGQDs is measurable in real

¹ Chen and Wu contributed equally this work.

time, and its quenching is very sensitive to the concentration of Fe^{3+} ions in the system but not to other possible coexisting metal ions. The fluorescence quenching mechanism of Fe^{3+} ions to BGQDs is studied and explained based on electrochemical voltammetry and DFT simulations. The analytical signal, which is defined as F_0/F , where F_0 and F are the fluorescence intensities of the BGQDs before and after interaction with Fe^{3+} ions, respectively, displays a good linear relationship in the Fe^{3+} ion concentration range of 0.01–100 μM with a correlation coefficient of 0.999 and a limit of detection (LOD) of $\sim(0.005 \pm 0.001)$ μM . The LOD value is much lower than the water quality standards for Fe^{3+} ions (0.3 ppm, ~ 5.36 μM) in drinking water suggested by the WHO (World Health Organization) and EPA (U.S. Environmental Protection Agency), implying that this method has great potential for applications in real sample assays. For example, the determination of the Fe^{3+} ion levels in three water samples (tap water, groundwater, and lake water) gives approximately the same results as those determined by the EPA-recommended AAS (atomic adsorption spectroscopy) method.

Graphical abstract

Keywords

Electrochemical synthesis; Boron-doped graphene quantum dots; Fluorescence quenching; Water sample; Fe^{3+} ion detection.

1. Introduction

Iron is a metal of biological, clinical, environmental, and industrial importance. Fe^{3+} ion is one of the most essential transition metal ions in biological system and it plays indispensable and versatile roles in many physiological and pathological processes, such as oxygen uptake and transport, cellular metabolism, enzyme catalysis, electron transfer process, and DNA and RNA synthesis [1-4]. However, both an excess and a deficiency of Fe^{3+} ions in the human body can induce serious disorders. A deficiency of Fe^{3+} ions can affect the synthesis of hemoglobin, causing anemia and limiting oxygen delivery to cells, resulting in fatigue, poor work performance, and decreased immunity [5,6]. Conversely, excess amounts of Fe^{3+} ions in a living cell can catalyze the production of reactive oxygen species (ROS) via the Fenton reaction, which can damage lipids, nucleic acids, proteins, and other cellular components [7]. These damages have been associated with the development of severe diseases including cancers, hepatitis, hemochromatosis, and organ dysfunction [8]. Moreover, recent studies have revealed that excess amounts of Fe^{3+} ions are also involved in neurodegenerative diseases, such as Parkinson's disease, Huntington's disease, and Alzheimer's disease [6,9].

Fe^{3+} ions also play an important role in natural aquatic systems and seawater [10,11]. For example, the aggregates of iron and humic substances (HSs) absorb trace metals, and Fe^{3+} ions can bind organic matter to clays [10], thus affecting the water environments. A fundamental study in oceanography based on the oligonutrient role of Fe^{3+} ions indicated that the Fe^{3+} ion is possibly the limiting agent for primary productivity (algal growth) in vast areas of the ocean [11]. Thus, low iron concentrations, which are consistently found in HNLC (High-Nutrient Low-Chlorophyll) areas, can impact the regulation of the global climate via a reduction in the ability of the oceanic phytoplankton to sink atmospheric CO_2 and export a part of it to the deep ocean in a process referred to as the biological pump [11].

The considerable importance of Fe^{3+} ions in biological and environmental systems has led to increasing interest in the development of selective techniques for the determination of Fe^{3+} ions. Various analytical techniques, such as atomic adsorption spectroscopy (AAS), which is the method recommended by the U.S. Environmental Protection Agency (EPA) for Fe^{3+} ion assays [12], inductively

coupled plasma atomic emission or mass spectroscopy (ICP-AES, ICP-MS) [13], spectrophotometry [14], and voltammetry [11], have been developed for sensitive Fe^{3+} ion determination, but several other metal ions have been shown to interfere with the analysis, necessitating complicated pretreatment procedures and sophisticated instrumentation.

Recently, fluorescent techniques have been widely investigated for the selective detection of Fe^{3+} ions in biological systems because of their ability to provide a simple, sensitive, and selective method for monitoring without the need for any pretreatment of the sample; these techniques also have the advantages of spatial and temporal resolution [15-17]. Rhodamine derivatives are extensively employed as fluorescence probes for the selective recognition of Fe^{3+} ions based on the Fe^{3+} -induced transformation from the spirocyclic form (colorless and nonfluorescent) to the ring-opened amide form (pink and highly fluorescent) [18,19]. However, the fluorescence of these probes is generally irreversible because of their reaction-based nature. Moreover, they intrinsically suffer from poor solubility and photostability, which hinders their potential in long-term biological imaging because of fast photobleaching. Wang et al. have coupled these organic molecules to PEG (polyethylene glycol)- Fe_3O_4 nanoparticles [20,21]; and Guo et al. have functionalized the rhodamine derivative (N-(rhodamine B) lactam-ethylenediamine, RBD) to graphene quantum dots (RBD-GQDs) to resolve the low solubility and improve the detection sensitivity [22]. In addition to the organic dye probes, many other fluorescence probes have been developed for Fe^{3+} ion assays, including fluorescent metal nanoclusters, semiconductor quantum dots (QDs), fluorescent metal organic frameworks [23-25], etc. Although these probes often provide admirable selectivity, their preparation usually requires specialized synthesis skills and complicated purification procedures. Thus, it has remained a challenge to create a fluorescence probe that possesses the required photostability, biocompatibility, selectivity, and sensitivity for Fe^{3+} ion levels below 0.3 ppm ($\sim 5.36 \mu\text{M}$), which is the maximum allowable level of Fe^{3+} ions in drinking water recommended by EPA [12] and WHO (World Health Organization) [26].

Graphene quantum dots (GQDs) have recently received tremendous interest due to their outstanding properties in terms of chemical stability, good dispersibility in water, stable photoluminescence (PL),

and long-term resistance to photobleaching, making them perfect alternatives to organic dyes and semiconductor quantum dots as fluorescent probes [27]. Various GQD-based fluorescent probes for the detection of metal ions (Cu^{2+} or Pb^{2+}) [28], small organic molecules (2,4,6-trinitrotoluene or paranitrophenol) [29,30], and biomaterials (pyrocatechol, human immunoglobulin G, hematin, or protein kinase) [31–33] have been explored. Despite the highly anticipated potential of these nanostructures, there is a need to develop GQDs with longer-wavelength emission and higher quantum yields (QYs), which are critically needed in biological imaging and environmental analysis as a result of the reduced background autofluorescence and improved signal-to-noise ratio.

Doping GQDs with heteroatoms can effectively modulate their intrinsic properties (such as electronic characteristics, optical properties, surface and local chemical features, and QYs), thereby producing new phenomena and unexpected properties for practical applications, which were proven through theoretical calculations and detailed experiments. For example, nitrogen-doped GQDs (NGQDs) showed efficient electrocatalytic activities for the reduction of oxygen [34], H_2O_2 [35] and 2,4,6-trinitrotoluene [30], and higher QY values for cellular and deep-tissue imaging [36]. Sulfur-doped graphene quantum dots (SGQDs) showed a stable blue-green fluorescence, drastically improved electronic properties, and increased surface chemical reactivity (compared with GQDs) [37]. Boron-doped GQDs (BGQDs) gave rise to rich fluorescence as a result of their specific interaction with the surrounding media [38].

The N atom (electron doping) and B atom (hole doping) are comparable in atomic size to the carbon atom, therefore, chemical doping of N or B atoms into the framework of GQDs would seem to be straightforward. Moreover, both N and B doping effectively tune the electronic nature of GQDs due to a difference in electronegativity between C (2.55) and N (3.04) or B (2.04) that appears to be large enough to result in significant charge transfer in C–N or C–B composites. Compared to the more established methods for the synthesis of NGQDs [34,39,40], we are aware that attempts made to synthesize BGQDs and the development of BGQD applications remain inchoate, which might be attributed to the limitation in synthesis methods and the difficulties in constructing an effective BGQD-based sensing platform. The BGQDs are usually synthesized by cutting boron-doped graphene using a

hydrothermal approach [41] or arc-discharge in gas phase [42]. These approaches may be limited by low QYs and product yields, and the required use of an organic solvent and strong oxidant. Considering the importance of doping the GQDs with holes, a facile route to synthesize high-quality BGQDs is still highly desired.

This work proposes a facile method to synthesize the BGQDs by simply electrolyzing a graphite rod in borax aqueous solution at a constant voltage (3 V), and using the BGQDs as a fluorescent probe to determine the presence of Fe^{3+} ions in water samples. The BGQDs emit a strong fluorescence under excitation of 300–440 nm with a dependence on the excitation wavelength and B-doping amount. We discover that this fluorescence can be efficiently quenched by the presence of Fe^{3+} ions and not by many other coexisting metal ions. The change in the fluorescence intensity of the BGQDs can be monitored in real time and is directly related to the amount of Fe^{3+} ions, which establishes the basis of measuring the level of Fe^{3+} ions in water. The observation that coexisting metal ions do not quench the fluorescence of the BGQDs shows that this assay is selective for Fe^{3+} ions, and there is no need to separate Fe^{3+} ions from samples by complicated pretreatments prior to the assay. In addition, this assay is label free, fairly simple, and inexpensive, supporting its potential application in real environmental samples.

2. Materials and methods

2.1. Synthesis of BGQDs

BGQDs were synthesized by a constant potential electrolysis method, which was performed on a CHI 660B electrochemical workstation (CH Instruments) in a two-compartment two-electrode cell with a sample volume of ~20 mL. The high purity graphite rod (99.9%, 3 mm in diameter, Shanghai Carbon Co. Ltd.) was used as an anode and was put into an aqueous 0.1 M borax (99.5%, Sigma-Aldrich) solution (adjusted to pH ~7); the length of the graphite rod below the solution level was approximately 10 cm. A Pt sheet (1.5 cm × 1.5 cm) was used as a cathode. The voltage between the anode and cathode was controlled at 3 V. After 2 h of electrolysis, BGQDs were collected by filtering the resultant solution using a 0.22 μm microporous nylon membrane to remove the precipitated graphite oxide and graphite

particles. Then, the obtained pale-yellow solution was dialyzed in deionized water in a dialysis bag (retained molecular weight of 3500 Da) for 48 h to remove the electrolyte of borax; the deionized water was changed every 12 h.

2.2. Characterizations

TEM images of the BGQDs were recorded on a JEOL JEM-2100F TEM operating at an accelerating voltage of 200 kV. The TEM samples were prepared by drying a droplet of the BGQDs solution on a carbon film-support grid. The surface characteristics of the BGQDs were examined by XPS, which was measured with an ESCALAB 250 XPS spectrometer (VG Scientifics) using a monochromatic Al K α line at 1486.6 eV. Binding energies were calibrated with respect to the C 1s peak at 284.6 eV. Peak fit analysis was performed using the XPS PEAK program (V4.0). XRD patterns of the BGQDs were recorded on a Rigaku/Max-3A X-ray diffractometer with Cu K α radiation ($\lambda = 0.15418$ nm). Raman spectra were recorded on a Labram HR 800 microspectrometer (Jobin Yvon) with an excitation source of 514 nm. UV-vis spectra were measured on a Cary 5000 UV-vis-NIR spectrometer (Varian). The FTIR transmission spectra were recorded at a resolution of 4 cm $^{-1}$ using a Nexus 670 FTIR spectrophotometer (Nicolet Instruments) for samples placed on a KBr disk. Fluorescence and photoluminescence (PL, under an excitation wavelength of 360 nm) spectra were collected using a FluoroSENS fluorescence spectrophotometer (Gilden Photonics) equipped with a Xenon lamp excitation source. The distribution and mean sizes of the BGQDs were analyzed by a BI-200SM dynamic light scattering instrument (DLS; Brookhaven Instruments).

2.3. Fe^{3+} -induced fluorescence quenching of BGQDs

For evaluating the fluorescence quenching ability of Fe^{3+} ions for the BGQDs, the fluorescence of the BGQDs in the presence of various concentrations of Fe^{3+} ions (The stock solution of Fe^{3+} ions was prepared from $FeCl_3 \cdot 6H_2O$ using 0.1 M HCl) was recorded under excitation of 380 nm. In a typical experiment, 15 μ g of the BGQDs was dispersed in 1 mL phosphate buffer (PBS, 0.1 M, pH 7.4), and the

fluorescence signal was recorded. Then, various concentrations of Fe^{3+} ions (0.01–100 μM) were added to the system and allowed to interact with the BGQDs. After a stable fluorescence of the solution was attained (usually within 5 min), the fluorescence signal was recorded and analyzed against the concentrations of Fe^{3+} ions.

For evaluating the selectivity of the fluorescence quenching of the Fe^{3+} ions to the BGQDs, several other possible co-existed metal ions such as Ni^{2+} , Fe^{2+} , Hg^{2+} , Na^+ , Mg^{2+} , Ag^+ , Pb^{2+} , Ca^{2+} , Zn^{2+} , K^+ , Cd^{2+} , and Co^{2+} ions were randomly selected as the potential interference to quench the fluorescence of BGQDs in the same procedure as that for Fe^{3+} ions. Ni^{2+} , Fe^{2+} , Hg^{2+} , Na^+ , Mg^{2+} , Ag^+ , Pb^{2+} , Ca^{2+} , Zn^{2+} , K^+ , Cd^{2+} , and Co^{2+} ion solutions were prepared from NiCl_2 , FeSO_4 , $\text{Hg}(\text{NO}_3)_2$, NaCl , MgCl_2 , AgNO_3 , $\text{Pb}(\text{Ac})_2$, CaCl_2 , ZnCl_2 , KCl , CdCl_2 , and CoCl_2 , respectively.

2.4. Electrochemical measurements

The redox characteristics of the BGQDs were studied by cyclic voltammetry, which was performed on an Autolab PGSTAT302N electrochemical station (Metrohm) using a glassy carbon (GC, 3 mm in diameter, CH Instruments) electrode. A Pt sheet (1.5 cm $\times$ 1.5 cm) and a saturated calomel electrode (SCE) were used as the counter and reference electrodes, respectively. All electrochemical measurements were performed at room temperature (22 ± 1 $^\circ\text{C}$). Prior to use, the GC electrode was sequentially polished with metallographic abrasive paper (no. 6) and slurries of 0.3 and 0.05 μm alumina to create a mirror finish. It was then sonicated for approximately 1 min in absolute ethanol and then double-distilled water to remove traces of alumina grit from its surface. After the electrode was rinsed thoroughly with double-distilled water and dried at ambient temperature, 15 μL of the BGQDs aqueous suspension (200 $\mu\text{g mL}^{-1}$) was cast onto the surface of the electrode with a microsyringe (the mass of the BGQDs loaded on the electrode surface was ~ 3 μg), and the solvent (water) was allowed to evaporate in air at ambient temperature. The redox features of the BGQDs were studied in freshly dried dimethylformamide (DMF) solution containing 0.1 M Bu_4NBF_4 (tetrabutylammonium tetrafluoroborate, from Sahn Chemical Technology) as the electrolyte.

2.5. Calculation details of the interaction of Fe^{3+} ion with BGQDs

The interaction of Fe^{3+} ions with the BGQDs was theoretically calculated based on DFT using Gaussian 03 (revision B.03; Gaussian Inc., Pittsburgh, PA, 2003) [43]. The geometric optimizations were performed using a DFT method with Becke's hybrid three-parameter nonlocal exchange functional combined with the Lee-Yang-Parr gradient-corrected correlation functional (B3LYP). The 6-31G (d, p) basis set was used for all elements. For these calculations, we constructed a model containing 28 hexagonal rings with delocalized π electrons (C77BH24) for BGQDs. The carbon atoms at the edge of the graphene were terminated with hydrogen atoms. For simulating Fe^{3+} ion adsorption, we placed Fe^{3+} ions near the BGQD plane at a distance of 3 Å and then optimized the geometric structures of the adsorbed Fe^{3+} ions at the surface of the BGQDs. The value of E_{ad} was obtained from frequency analysis after the initial structure optimization. The energy of the isolated system is the sum of the energies of the BGQDs and the isolated adsorbed molecule. Thus, a negative value of E_{ad} indicates that the adsorption process is energetically favorable. The effects of the interaction of Fe^{3+} ions with the BGQDs on the energy gap between the LUMO and HOMO energy levels of BGQDs was also evaluated using the calculation.

3. Results and discussion

3.1. Synthesis and characterization of BGQDs

For determination of the Fe^{3+} ion levels in water samples using BGQDs as fluorescent probes, we synthesized BGQDs by a constant potential electrolysis method, which was performed in an aqueous 0.1 M borax (99.5%) solution (adjusted to pH ~7). A high purity graphite rod (99.9%, 3 mm in diameter) was used as the anode and a Pt sheet (1.5 cm × 1.5 cm) was used as the cathode. The voltage between the anode and cathode was controlled at 3 V (the current flowing between the anode and cathode was observed to be ~0.13 mA). Assuming the Pt cathode was pinned by the proton reduction reaction at pH 7, then its potential is around -0.4 V versus the normal hydrogen electrode (NHE), and the graphite

anode would have to be approximately +2.6 V versus NHE to maintain the cell voltage at 3 V. This voltage (+2.6 V) is high enough to drive the electrolyte ions into the graphene layers and to oxidize the C–C bonds to form B–C bonds and BGQDs [34,38]. As the electrolysis time increased, the color of the solution in the anode area was observed to change from colorless to pale yellow and finally to brown. At the same time, a large amount of gas evolved from the cathode, indicating the reduction of protons to hydrogen, and the oxidation and dissolution of the graphite anode, leading to the formation of the BGQDs. After 2 h of electrolysis, BGQDs were collected by filtering the resultant solution using a 0.22 μm microporous nylon membrane to remove the precipitated graphite oxide and graphite particles. Then, the obtained pale-yellow solution was dialyzed over deionized water in a dialysis bag (retained molecular weight of 3500 Da) for 48 h to remove the borax electrolyte (the deionized water was changed every 12 h). The obtained BGQD solution remains homogeneous with a pale yellow color for several months (at least a span of four months in our experiments) at ambient temperature without any perceptible aggregation and color change (Fig. S1), which could be further verified by the almost unchanged PL spectra after four months (Fig. S2). This feature is crucial because the synthesized BGQDs must be water-soluble and stable in the ambient environment for their practical use in sensing applications. The high dispersion and physical stability of the BGQD solution is probably due to the oxygen-containing groups present at the surface and edges of the BGQDs, as evidenced by the FTIR spectrum (see the discussion presented below).

The concentration of the aqueous BGQD solution was found by taking a known volume of the aqueous BGQD solution, evaporating it to dryness, and weighing the dry mass of the BGQDs, which gave a concentration of 15 $\mu\text{g mL}^{-1}$. Approximately 320 μg of BGQDs can be obtained by passing 1 coulomb of charge, implying that this electrolyzing method can be used for the high-yield synthesis of BGQDs.

Transmission electron microscopic (TEM) and high-resolution TEM (HRTEM) images show that the BGQDs are well monodispersed (Fig. 1A) and have a relatively narrow size distribution ranging from 3 to 6 nm with an average diameter of ~ 4.5 nm (Fig. 1B), which is in good agreement with the results

obtained by dynamic light scattering (DLS) analysis (Fig. S3). The HRTEM image shows the high crystallinity of the BGQDs with a lattice spacing of 0.241 nm (Fig. 1C), which corresponds to the lattice constant in the (1220) plane of graphitic (sp^2) carbon. However, this lattice constant is slightly smaller than that of pristine graphite (0.242 nm) [44], implying that the boron atoms have been doped into the conjugated carbon backbone, leading to somewhat disordered structures.

<Fig. 1.>

Further evidence of the B-doping in the graphene lattice is provided by the Raman and FTIR spectra. The Raman spectrum of the BGQDs exhibits two peaks at 1347 and 1601 cm^{-1} corresponding to the well-defined D and G bands (curve a, Fig. 1D), respectively. The ratio of I_D/I_G in Raman spectra of carbon materials is usually used to evaluate the disorder in the graphene structures because the G band is associated with the degree of graphitization, while the D band is associated with structural defects and partially disordered structures of the sp^2 domains. The value of the I_D/I_G ratio of the BGQDs is estimated to be ~0.95 (curve a, Fig. 1D), which is higher the value for pristine graphene (~0.80, curve b, Fig. 1D), suggesting a breakage of the hexagonal symmetry of the graphene [42], most presumably caused by the B-doping. These results indicated that BGQDs have been synthesized by the electrolysis method.

In the FTIR spectrum of the BGQDs (Fig. 1E), we can observe the absorption peaks corresponding to the stretching mode of O–H (~3442 cm^{-1}), stretching vibration of C–O (~1578 cm^{-1}), sp^2 hybridized C–C (~1431 cm^{-1}), C–OH groups (~1338 cm^{-1}), and the deformation mode of C–O–C in epoxide moieties (~1129 cm^{-1}) [45,46], suggesting that there is an abundance of the oxygen-containing groups present at the surface and edges of the BGQDs. The appearance of the peak at ~1405 cm^{-1} , which corresponds to the asymmetric stretching vibration mode of the B–O bond [47], further confirms the incorporation of boron during the synthesis of the BGQDs.

The XRD pattern of the BGQDs shows a diffraction peak (2θ) at ~ 24.7 degrees (curve a, Fig. 1F), corresponding to the (002) plane of graphite. In contrast to the sharp peak of the pristine graphite (curve b, Fig. 1F), this diffraction peak is broader, indicating that the oxidation process during the synthesis of BGQDs introduced more active sites on the surface of BGQDs, as was indicated by the FTIR results and can be further confirmed by XPS (see the discussion presented below). Moreover, the XRD peak of the BGQDs shifts to a lower degree compared with that of pristine graphite ($\sim 26^\circ$), implying that the BGQDs have a larger interlayer spacing (calculated to be ~ 0.360 nm) than pristine graphene does (0.342 nm [41]). The increased interlayer spacing can be attributed to the weak π - π stacking of the graphene with more oxygen-containing functional groups at the edges of the layer of BGQDs. It is also observed that the BGQDs do not show any diffractions at $\sim 10^\circ$, a characteristic feature of graphene oxide (GO), suggesting that the XRD characteristics of BGQDs are different from GO, although both of them are rich in oxygen-containing groups.

The detailed surface composition and bonding information of the doped B atom in the BGQDs can be studied by X-ray photoelectron spectroscopic (XPS) spectra, which show the presence of B, C, and O elements with atomic percentages of ~ 3.2 , 60.1 , and 36.7% , and corresponding B1s, C1s, and O1s peaks at ~ 191 , 284 , and 531 eV (Fig. 1G), respectively. This further confirms the incorporation of B atoms into the graphene by electrolysis of the graphite in borax solution. The B1s peak appears at a higher binding energy (~ 191 eV) compared with that reported for pure boron (~ 187 eV [41]), suggesting that B atoms from the borax have been bonded to C atoms in the sp^2 carbon network. The high-resolution XPS peak of C1s indicates the presence of the C-B (~ 283.9 eV [48]), C=C (~ 284.6 eV), C-O (~ 285.3 eV), C=O (~ 286.5 eV), O-C=O (~ 288.5 eV) bonds [46,47] (Fig. 1H). These results suggest that there is a large amount of oxygen-containing groups present at the surface of the BGQDs, agreeing with the results from the FTIR spectrum (Fig. 1E). In the high-resolution XPS spectrum, the B1s peak is deconvoluted into three peaks at ~ 189.8 , 190.6 , and 191.3 eV, corresponding to the B bonding formed in C-B (~ 189.8 eV), BC_2O (~ 190.6 eV), and BCO_2 (~ 191.3 eV) [49] (Fig. 1H), respectively. The average amounts of the C-B, BC_2O , and BCO_2 components are $\sim (24 \pm 3.6)$, (55 ± 6.1) , and $(21 \pm 4.7)\%$

(averaged from five different BGQD samples, which were synthesized using the same procedure), respectively. These results further indicate the doping of the B atom into graphene and formation of the BGQDs.

The B doping amount in BGQDs can be tuned by simply changing the concentration of the borax used in the electrolyzing synthesis of BGQDs. Our results based on the XPS analysis show that the B amount doped into the BGQDs is ~1.8, 3.2, 4.3, and 5.1% (in B atomic percentage), respectively, when borax concentrations of 0.05, 0.1, 0.15, and 0.2 M are used.

3.2. Adsorption and photoluminescence (PL) characteristics

After the synthesis of BGQDs was confirmed, their adsorption and photoluminescence (PL) emission characteristics were studied. The UV-vis spectrum of the BGQDs is shown in Fig. 2A. A typical adsorption peak at ~230 nm is observed and is assigned to the $\pi-\pi^*$ transition of aromatic sp^2 domains [50], which dominate the BGQD structure. Two weak adsorption peaks at ~302 and 370 nm are also observed and are assigned as $n\rightarrow\pi^*$ transitions of electrons in the B–C bond [33], which is indicative of B doping in graphene. The absorption edge may correspond to $n\rightarrow\pi^*$ transitions of nonbonding electrons of oxygen atoms involved in C=O and/or C–O bonds [51]. It has been reported that isolated sp^2 -hybridized clusters with a size of ~3 nm within the carbon-oxygen matrix can generate band gaps consistent with blue emission due to the localization of the electron-hole pairs [30,52]. To check whether our BGQDs emit a blue PL emission, the BGQD solution was irradiated at 365 nm (16 W). It is interesting to observe that the BGQDs emit a blue PL (Fig. S4), while its solution shows no noticeable PL without UV irradiation and remains pale-yellow, transparent, and clear under daylight (Fig. S4). This suggests that both the size and surface features make important contributions to the observed blue PL emission from BGQDs.

<Fig. 2.>

An excitation wavelength-dependent PL emission was observed with the BGQDs as depicted in Fig. 2B, which shows two prominent PL emission regions at ~ 465 and 530 nm. Upon 300–340 nm irradiation, the first PL peak (at ~ 465 nm) dominates the second peak (at ~ 530 nm), and its intensity (the peak at 465 nm) increases with the wavelength of excitation. When the excitation wavelength increases to 360 nm, the second PL peak (at 530 nm) dominates the first peak (at 465 nm), and its intensity decreases rapidly when the excitation wavelength is further increased to 440 nm; this is characteristic of the PL for most luminescent carbon nanoparticles. As in the case of most luminescent carbon dots and graphene quantum dots, this excitation-dependent fluorescence behavior of the BGQDs reflects the effect of particles of different sizes and a distribution of different surface states [50].

The PL intensity of the BGQDs at extreme pH or high ionic strength in solution was investigated to evaluate their stability under extreme conditions. BGQDs displayed stable PL even under extreme pH conditions because they exhibited a strong PL activity in the range of pH 3 to 11 (curve a, Fig. 2C). The effect of the ionic strength on the PL activity was evaluated by recording the PL signal in solutions containing different concentrations of NaCl (0.01 to 1 M). The PL intensity remains constant with an increase in the ionic strength (curve b, Fig. 2C). These results suggest that BGQDs are highly stable even under extreme conditions.

The photoluminescence excitation (PLE) spectrum, which reveals the excitation wavelengths for local maximum PL emissions at the peak wavelengths depicted in PL spectra (Fig. 2B), of the BGQDs was also studied. The PLE spectrum depicted in Fig. 2D was monitored at the wavelengths for the maximal PL emissions of 530 nm from the BGQDs. The PLE spectrum shows two peaks at ~ 270 and 360 nm. The peak at ~ 270 nm (4.60 eV) is believed to result from the $n \rightarrow \sigma^*$ transitions [50], while the peak at ~ 360 nm (3.45 eV) is ascribed to the $n \rightarrow \pi^*$ transitions [50]. These peaks have been reported for GO in the sp^3 domains associated with hydroxyls, amines, and ethers. The energy difference between the σ^* and π^* orbitals is 1.15 eV, which is within the value expected for triplet carbenes (<1.5 eV) [51], suggesting that the assignment of the two transitions is reasonable. Because the two transitions are closely related to the emitted PL of the BGQDs, we can conclude that the observed blue PL emission

resulted from the irradiation decay of the activated electrons from the LUMO to HOMO. Therefore, we have reconfirmed the synthesis of BGQDs from their spectroscopic and optical characteristics.

<Fig. 3.>

The PL characteristics of the synthesized BGQDs also depend on the B doping amount. The intensities of the PL and PLE also increase with the B doping amount (curves a–d for the PL spectra, and curves a'–d' for the PLE spectra, Fig. 3). In the PLE, the peaks are caused by the transitions of the n electrons in the B–C bonds (the $n \rightarrow \sigma^*$ transitions for the peak at ~ 270 nm, and $n \rightarrow \pi^*$ transitions for the peak at ~ 360 nm). In the PL, the peak (~ 530 nm) is caused by the radiative $\pi^* \rightarrow n$ transitions. The increased amount of the B doping results in an increase the amount of B–C bonds; thus, the PLE and PL peak intensities increase.

The quantum yields (QYs) of the BGQDs with different B doping amounts were estimated to be approximately 3.6, 5.2, 7.6, and 9.3% for BGQDs with the B doping amounts of ~ 1.8 , 3.2, 4.3, and 5.1%, respectively, by comparing the integrated photoluminescence intensities (excited at 380 nm) and the absorbance values (at 380 nm) using rhodamine 6G in ethanol solution (QY = 95%) as a reference.

3.3. Fe^{3+} -induced fluorescence quenching of BGQDs

We then studied the Fe^{3+} -induced fluorescence quenching of BGQDs. The BGQDs exhibit a strong characteristic fluorescence peak at ~ 530 nm under an excitation of 380 nm (curve a, Fig. 4A). Upon the addition of Fe^{3+} ions (100 μ M), we observed an almost complete quenching of the fluorescence of the BGQDs (curve b, Fig. 4A), revealing that the fluorescence intensity of the BGQDs is sensitive to the presence of Fe^{3+} ions. It should be noted that the fluorescence of the Fe^{3+} ion itself does not interfere with the observation of the fluorescence features of BGQDs because the fluorescence intensity of Fe^{3+} ion alone is very low and is characterized by a flat curve (curve c, Fig. 4A) under the same excitation conditions as those for the BGQDs.

<Fig. 4.>

Kinetic studies show that the fluorescence quenching of Fe^{3+} ions is relatively fast; the fluorescence of the BGQDs was almost completely quenched within 1 min after adding the Fe^{3+} ions into the system (Fig. 4B, C). However, no time-dependent fluorescence changes were observed for the BGQDs themselves in solution in the absence of Fe^{3+} ions, implying that the time-dependent changes depicted in Figs. 4B and 4C originate from the Fe^{3+} ion-induced fluorescence quenching of BGQDs. The promising quenching ability of Fe^{3+} ions may arise from the strong adsorption affinity and highly efficient energy transfer between BGQDs and Fe^{3+} ions.

To further study the mechanism of the Fe^{3+} -induced fluorescence quenching of BGQDs, we performed DFT (Density Functional Theory) simulations based on Gaussian 03 [43] to understand the interaction of the Fe^{3+} ions with BGQDs at the microscopic level. The geometric optimizations were performed using a Becke's hybrid three-parameter nonlocal exchange functional combined with the Lee-Yang-Parr gradient-corrected correlation functional (B3LYP). The 6-31G (d, p) basis set was used for all elements. For these calculations, we constructed a model containing 28 hexagonal rings with delocalized π electrons (C77BH24) for the BGQDs. The carbon atoms at the edge of the graphene were terminated with hydrogen atoms (please refer to the Experimental section for the calculation details). Mulliken charge density analysis indicates that carbon atoms in pure GQDs are almost neutral, except for those near the edge (H-terminated), and show very weak negative charges due to hydrogen bonding. In BGQDs, a significant positive charge is located on the doped boron atom (Fig. 5A) because carbon has a larger electronegativity and stronger electron-accepting ability than boron. This positively charged boron atom breaks the electroneutrality of sp^2 carbon to create charged sites favorable for the interaction of Fe^{3+} ions with BGQDs. The adsorption of the Fe^{3+} ion causes the redistribution of the charges on the BGQD plane. The Fe^{3+} ions are located ~ 1.63 Å over the BGQD plane, and the distance of the adsorbed Fe^{3+} ion to the B, C1, C2, C3, C4, and C5 atom is $\sim 2.23, 2.18, 2.21, 2.12, 2.21,$ and 2.18 Å (Fig. 5B, and

the inset), respectively. The adsorption energy (E_{ad}), which was defined as the zero-point energy (ZPE) difference between the total system, including the adsorbed molecules, and the isolated system of Fe^{3+} ions on BGQDs was calculated to be about -1.65 eV based on frequency analysis after the initial structure optimization. The energy of the isolated system is the sum of the energies of the BGQDs and the isolated adsorbed molecule. Thus, a negative value of E_{ad} indicates that the adsorption process is energetically favorable. The amount of the charge transferred from the BGQDs to adsorbed Fe^{3+} ion is $\sim 2.1e^-$, implying a strong interaction between Fe^{3+} ion and BGQDs.

<Fig. 5.>

To understand efficient energy transfer between the BGQDs and Fe^{3+} ions, we estimated the values of the HOMO and LUMO of BGQDs through a combination of electrochemical measurements and UV-vis spectra. The HOMO and LUMO energy level and the onset potentials for the oxidation (ϕ_{ox}) and reduction (ϕ_{red}) of the BGQDs can be obtained by the following empirical formulas [25,54,55]:

$$E_{HOMO} \text{ (in eV)} = -(\phi_{ox} + 4.74) \quad (1)$$

$$E_{LUMO} \text{ (in eV)} = -(\phi_{red} + 4.74) \quad (2)$$

The ϕ_{ox} was determined to be ~ 0.34 V (vs. SCE) from the cyclic voltammogram of the BGQDs (Fig. S5). The corresponding E_{HOMO} was calculated to be -5.08 eV. The E_{LUMO} could not be estimated from eq 2 because of the irreversibility of the BGQD oxidation (Fig. S5). However, the E_{LUMO} can be estimated using eq 3 [55]:

$$E_{HOMO} = E_{LUMO} - E_g \quad (3)$$

where E_g is the HOMO–LUMO energy gap, which is estimated to be ~ 3.23 eV from the absorption edge in the UV-vis spectrum ($\lambda = 385$ nm, Fig. 2A). These values are very close to those obtained from DFT calculations, which show that the HOMO and LUMO values of BGQDs are -5.05 and -1.83 eV, respectively, with an E_g of 3.22 eV. The E_{HOMO} and E_{LUMO} of Fe^{3+} ion were calculated to be -4.90 and -2.22 eV, respectively. The calculated HOMO–LUMO energy gap (2.68 eV) is very close to the reported

crystal field stabilization energy (CFSE) of the Fe^{3+} ion (~ 2.70 eV) [56]. In the absence of Fe^{3+} ion, the BGQDs exhibited strong fluorescence (Fig. 5C). In the presence of Fe^{3+} ions, however, the Fe^{3+} ions can be adsorbed on the surface of BGQDs and induce the partial transfer of excited electrons from the BGQDs to the LUMO orbitals of the Fe^{3+} ion, thereby facilitating nonradiative electron/hole recombination processes; thus, the electron transition in the radiative forms (fluorescence emission) was restrained, causing the fluorescence quenching of the BGQDs (Fig. 5D).

The time-resolved fluorescence measurements, which were carried out on a time-correlated single photon counting (TCSPC) system, were applied to further check the charge transfer and exciton recombination process of the BGQDs in the presence and absence of Fe^{3+} ions. As shown in Fig. S6, the average fluorescence lifetime of pure BGQDs (curve a) is 6.9 ns, obtained using a deconvolution technique by IBH DAS6 analysis software, reflecting a fast exciton recombination process. After the addition of Fe^{3+} ions (curve b), the lifetime of the BGQDs decreased to 1.3 ns. The significantly reduced fluorescence lifetime indicates that dynamic quenching occurred, and further confirms the fast electron transfer process in the BGQDs- Fe^{3+} system.

3.4. Assay of Fe^{3+} ions in water samples

The strong fluorescence quenching behavior of Fe^{3+} ions and the quick fluorescence response of BGQDs to Fe^{3+} ions can be employed as a fluorescent probe for real-time assays of Fe^{3+} ions in water samples. To achieve this goal, we first studied the fluorescence changes of BGQDs in the presence of different concentrations of Fe^{3+} ions. As shown in Fig. 6A, the fluorescence intensity of the BGQDs decreases gradually with increasing Fe^{3+} ion concentration in solution (from 0.01 to 100 μM), implying that the fluorescence of BGQDs is sensitive to the Fe^{3+} ion concentration. We employed the value of F_0/F , where F_0 and F are the fluorescence intensities of the BGQDs before and after interaction with Fe^{3+} ions, respectively, as the analytical signal and plotted the relationship between F_0/F and the concentration of Fe^{3+} ions. To accurately determine the relationship between F_0/F and the concentration of Fe^{3+} ions, the observed fluorescence emission intensities were corrected for the inner filter effect due

to the absorbance of Fe^{3+} ions at the excitation and emission wavelengths as described by MacDonald et al [57] because the absorbance of Fe^{3+} ion may cause analytical errors [57,58].

$$F_{\text{corr}} = F_{\text{obs}} \times 10^{\frac{A_{\text{ex}} + A_{\text{ems}}}{2}} \quad (4)$$

where F_{corr} and F_{obs} are the corrected and observed fluorescence intensities, respectively, and A_{ex} and A_{ems} are the absorbance of the solution at the excitation (380 nm) and emission wavelengths (530 nm), respectively. The typical UV-vis spectra of the Fe^{3+} ions at the concentrations of 0.01–100 μM are depicted in Fig. S7, showing that the absorbance of Fe^{3+} ions at the excitation wavelength (380 nm) is ~ 0.09 at a Fe^{3+} ion concentration of 100 μM , which is the highest concentration used in this work. The absorbance of Fe^{3+} ions at the emission wavelength of 530 nm is negligible. After correction, F_0/F was plotted against the concentration of Fe^{3+} ion as shown in Fig. 6B. F_0/F increases with the concentration of Fe^{3+} ion and displays a good linear relationship in the concentration range of 0.01–100 μM with a correlation coefficient (R^2) of 0.999. The limit of detection (LOD) is estimated to be $\sim (0.005 \pm 0.001)$ μM based on an $3\delta/S$, where δ is the standard deviation of the blank signal and S is the slope of the linear calibration plot. This LOD value is much lower than the water quality standards for Fe^{3+} ions (0.3 ppm, $\sim 5.36 \mu\text{M}$) in drinking water suggested by the WHO and EPA [12,26]. It is also much lower than most of the previous reported assays for Fe^{3+} detection using other fluorescence probes. For example, Wang et al. [59] reported an LOD of 17.9 μM for Fe^{3+} ions using graphene oxide as the fluorescence probe; Weerasinghe et al. [60], Chai et al. [61], Chen et al. [62], Kagit et al. [16], and Yang et al. [63] reported detection limits for Fe^{3+} ions of 100, 20, 10, 4.8, and 5 μM , respectively, using rhodamine-based fluorescence probes.

<Fig. 6.>

The reproducibility of the BGQD fluorescence quenching method for the Fe^{3+} ion assay was evaluated by using a relative standard deviation (RSD) value of $\sim 3.2\%$, which was estimated from the

slopes of calibration plots for five different measurements, revealing the acceptable reproducibility of this method. The assay precision was examined by five determinations at a Fe^{3+} ion concentration of 10 μM . An RSD of $\sim 2.2\%$ is obtained in this case. These results mean that the method can be used as a sensing platform for detecting Fe^{3+} ions.

We then studied the selectivity of the fluorescence quenching of the BGQDs by Fe^{3+} ions because the complexity of the environmental water poses a great challenge to analytical methods for metal ion detection not only in the area of sensitivity but, more importantly, with regard to selectivity. The fluorescence intensity ratios (F/F_0 , F_0 and F are the fluorescence intensities of the BGQDs before and after the addition of different metal ions, such as Fe^{3+} , Ni^{2+} , Hg^{2+} , Fe^{2+} , Na^+ , Mg^{2+} , Ag^+ , Pb^{2+} , Ca^{2+} , Zn^{2+} , K^+ , Cd^{2+} , and Co^{2+} ions, 100 $\mu\text{g mL}^{-1}$) of the BGQDs (15 $\mu\text{g mL}^{-1}$) were analyzed upon adding different metal ions to the sensing system. As shown in Fig. 6C, only Fe^{3+} ions caused significant quenching of the fluorescence of the BGQDs, and the effect of the other tested metal ions was negligible, indicating the high selectivity of the BGQDs for Fe^{3+} ions in aqueous solution. Further experiments to understand the effect of coexisting metal ions on the quenched fluorescence intensity of the BGQDs by Fe^{3+} were also performed to show the anti-interference abilities of the BGQDs for detecting Fe^{3+} ions. These results highlight the high selectivity of the BGQD fluorescence probe for detecting Fe^{3+} ions and the great potential of BGQDs for real sample applications.

The reversibility of the BGQDs as a fluorescent probe to detect Fe^{3+} ions was studied. After the fluorescence of the BGQDs was quenched by Fe^{3+} ions (100 μM , curves a and b, Fig. S8), EDTA (ethylene diamine tetraacetic acid tetrasodium salt, 100 μM) was added to the mixture. We observed that the fluorescence intensity of the BGQDs was almost completely recovered (curve c, Fig. S8), implying that EDTA can coordinate readily with the adsorbed Fe^{3+} ions to form the more stable Fe-EDTA, and release the BGQDs. This result suggests that the BGQD fluorescent probe can be regenerated in a Fe^{3+} ion assay.

To explore the practical application of our method for Fe^{3+} ion detection in a real water sample, we studied three water samples, tap water, groundwater, and lake water. These samples were first

contaminated with 5, 25, and 50 μM Fe^{3+} ions, and then the recovery measurements were carried out. The average recoveries for the three different samples are 99.0, 100.4, and 99.4%, respectively, and the RSDs of the recoveries are 0.25, 1.26, and 0.84% (Table 1), respectively.

To confirm the accuracy of our assay, the amount of Fe^{3+} ions in those water samples was measured using AAS, which is the method recommended by the EPA for Fe^{3+} ion assay. For better comparison, the amount of Fe^{3+} ions in water sample determined by AAS was also presented in Table 1. It is shown the amounts of Fe^{3+} ions in water samples determined by our fluorescence method and the standard AAS method are essentially the same after considering the experimental errors, indicating that the BGQDs fluorescence quenching method has high accuracy in detecting the Fe^{3+} ions in real water samples and therefore has high potential for use as a sensing platform for practical Fe^{3+} ion assays.

4. Conclusions

In summary, a facile electrolyzing method has been reported to synthesize the boron-doped graphene quantum dots (BGQDs). The excitation wavelength- and B amount-dependent fluorescence characteristics of BGQDs have been studied. The fluorescence intensity of the BGQDs has been found to be very sensitive to the presence of Fe^{3+} ion in the system but not to other possible coexisting metal ions, showing the high selectivity of this Fe^{3+} ion assay and highlighting that there is no need for tedious procedures to separate Fe^{3+} ion from other metal ions prior to the assay. The analytical signal displays a good linear relationship in the Fe^{3+} ion concentration range of 0.01–100 μM with a correlation coefficient of 0.999 and a limit of detection (LOD) of $\sim(0.005 \pm 0.001)$ μM . The LOD value is much lower than the suggested water quality standards for Fe^{3+} ions (0.3 ppm, ~ 5.36 μM) in drinking water by WHO and EPA, implying that this method has the potential for applications in real water sample assays. The fluorescence quenching mechanism of Fe^{3+} ion to BGQDs has also been studied and explained based on electrochemical voltammetry and DFT simulations. The advantages of this method are that it is facile, sensitive, label-free, and inexpensive.

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Appendix A. Supplementary data

Characterization and electrochemical characteristics of the BGQDs, the UV-vis spectra of Fe^{3+} ions, and the reversibility of Fe^{3+} -induced fluorescence quenching of BGQDs.

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Table**Table 1. Determination of Fe³⁺ ions in three water samples using the BGQDs as a fluorescence probe**

Samples ^a	added amounts (μM)	measured amounts ^b (μM)	recovery (%)	RSD (%)	measured amounts ^c (μM)
tap water	5	4.98	99.6	1.48	4.98
	25	25.40	101.6		24.96
	50	49.33	98.7		49.90
ground water	5	5.03	100.6	1.04	4.95
	25	24.65	98.6		24.96
	50	50.06	100.1		49.85
lake water	5	4.95	99.0	1.15	4.96
	25	25.22	100.9		25.06
	50	49.39	98.8		50.12

^a The Fe³⁺ contents in the tap water, ground water, and lake water are $\sim(0.003 \pm 0.001)$, (0.024 ± 0.005) , and (0.015 ± 0.003) μM, respectively, measured by AAS. ^b Values determined by the BGQDs as fluorescence probe; it is an average value of the response of five measurements for each sample. ^c Values determined AAS; it is an average value of the response of five measurements for each sample.

Figure Captions

Fig. 1. Typical TEM (A) and HRTEM images (B, C) of BGQDs. (D, E, and F) Raman spectrum (D, curve a), FTIR spectrum (E), and XRD pattern (F, curve a) of the BGQDs. Curve b in (D) and (F) shows the Raman spectrum and XRD pattern, respectively, of the pristine graphene. The inset in (C) shows the presence of the B–O bond in NGQDs. (G) XPS spectrum of BGQDs. The inset shows the amplified XPS of B1s. (H and I) High-resolution XPS spectra of C1s (H) and B1s (I) in BGQDs, respectively, and the related curve-fitted components.

Fig. 2. (A) UV-vis spectrum of the BGQDs. (B) Dependence of PL spectra of BGQDs on excitation wavelength. (C) The effects of the solution pH (a) and ionic strength (b) on the fluorescence signals of the BGQDs. (D) PLE spectrum of BGQDs. The PLE spectrum was monitored at the wavelengths for the maximal PL emissions of 530 nm. The BGQDs were synthesized by electrolyzing the graphite rod in 0.1 M borax solution at 3 V.

Fig. 3. PL (a–d) and PLE (a'–d') spectra of BGQDs with the different B doping amount. The doped B amount of BGQDs is ~1.8 (a, a'), 3.2 (b, b'), 4.3 (c, c'), and 5.1 at% (d, d'), which was synthesized by electrolyzing the graphite rod in borax aqueous solution at 3 V at borax concentration of 0.05, 0.1, 0.15, and 0.2 M, respectively. The PLE spectrum was monitored at the wavelengths for the maximal PL emissions of 530 nm from BGQDs.

Fig. 4. (A) Fluorescence spectra of BGQDs alone (curve a, $15 \mu\text{g mL}^{-1}$) and in the presence of Fe^{3+} ion (curve b, $100 \mu\text{M}$). The curve (c) shows the fluorescence spectrum of Fe^{3+} ions ($100 \mu\text{M}$) alone. The excitation wavelength is 380 nm. (B) Fluorescence spectra of the BGQDs after it interacted with Fe^{3+} ions ($100 \mu\text{M}$) for (a–f) 0, 1, 5, 10, 15, and 20 min. (C) Time-dependent fluorescence signal of BGQDs based on the quenching time. Error bars were based on five measurements.

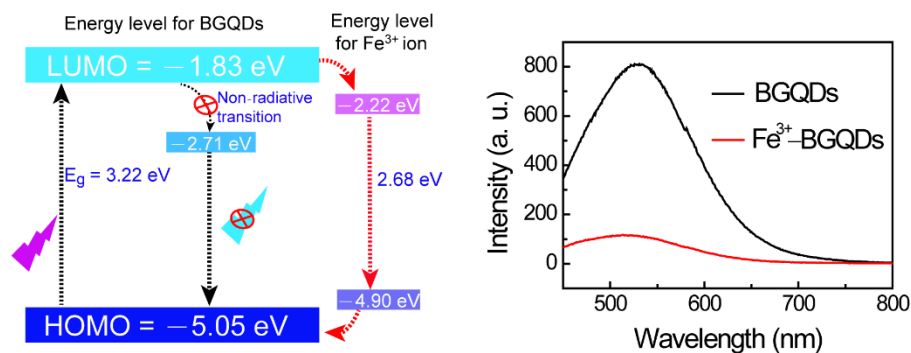
Fig. 5. (A) Calculated charge distribution on boron and each carbon atom in BGQD plane. The red and green colors represent negative and positive charges, respectively. (B) Optimized configurations of Fe^{3+} ion adsorbed on the surface of BGQDs. The inset shows the side view of the position of the adsorbed Fe^{3+} ion over the BGQD plane. (C, D) Diagram illustrating the fluorescence emission of BGQDs (C) and the fluorescence quenching of BGQDs after adsorption of Fe^{3+} ion on their surface (D). The values of the HOMO and LUMO for BGQD and Fe^{3+} ion were obtained by DFT calculations.

Fig. 6. (A) Fluorescence spectra of BGQDs in the presence of different concentrations of Fe^{3+} ions. The concentrations of Fe^{3+} ion are (a–s) 0, 0.01, 0.05, 0.1, 0.5, 1, 2, 4, 6, 8, 10, 15, 20, 25, 30, 40, 60, 80, 100 μM . The concentration of the BGQDs is $\sim 15 \mu\text{g mL}^{-1}$. The excitation wavelength is 380 nm. (B) Dependence of F_0/F on the Fe^{3+} ion concentration. F_0 and F are the fluorescence intensities of the BGQDs before and after interaction with Fe^{3+} ions, respectively. (C) Fluorescence intensity ratios (F/F_0) of the BGQDs in the presence and absence of different metal ions (100 μM) listed from left to right: blank, Fe^{3+} , Ni^{2+} , Hg^{2+} , Fe^{2+} , Na^+ , Mg^{2+} , Ag^+ , Pb^{2+} , Ca^{2+} , Zn^{2+} , K^+ , Cd^{2+} , and Co^{2+} ions. The error bars represent the standard deviations based on the independent measurements.

Highlights

- A facile electrolyzing method to synthesize boron-doped graphene quantum dots (BGQDs) is reported.

- The excitation wavelength- and B amount-dependent fluorescence characteristics of BQGDs are studied.
- BQGDs can be used as a fluorescent probe to determine Fe^{3+} ion levels in water samples.
- It can determine Fe^{3+} ion concentration as low as $\sim(0.005 \pm 0.001) \mu\text{M}$ with linear range of 0.01–100 μM .



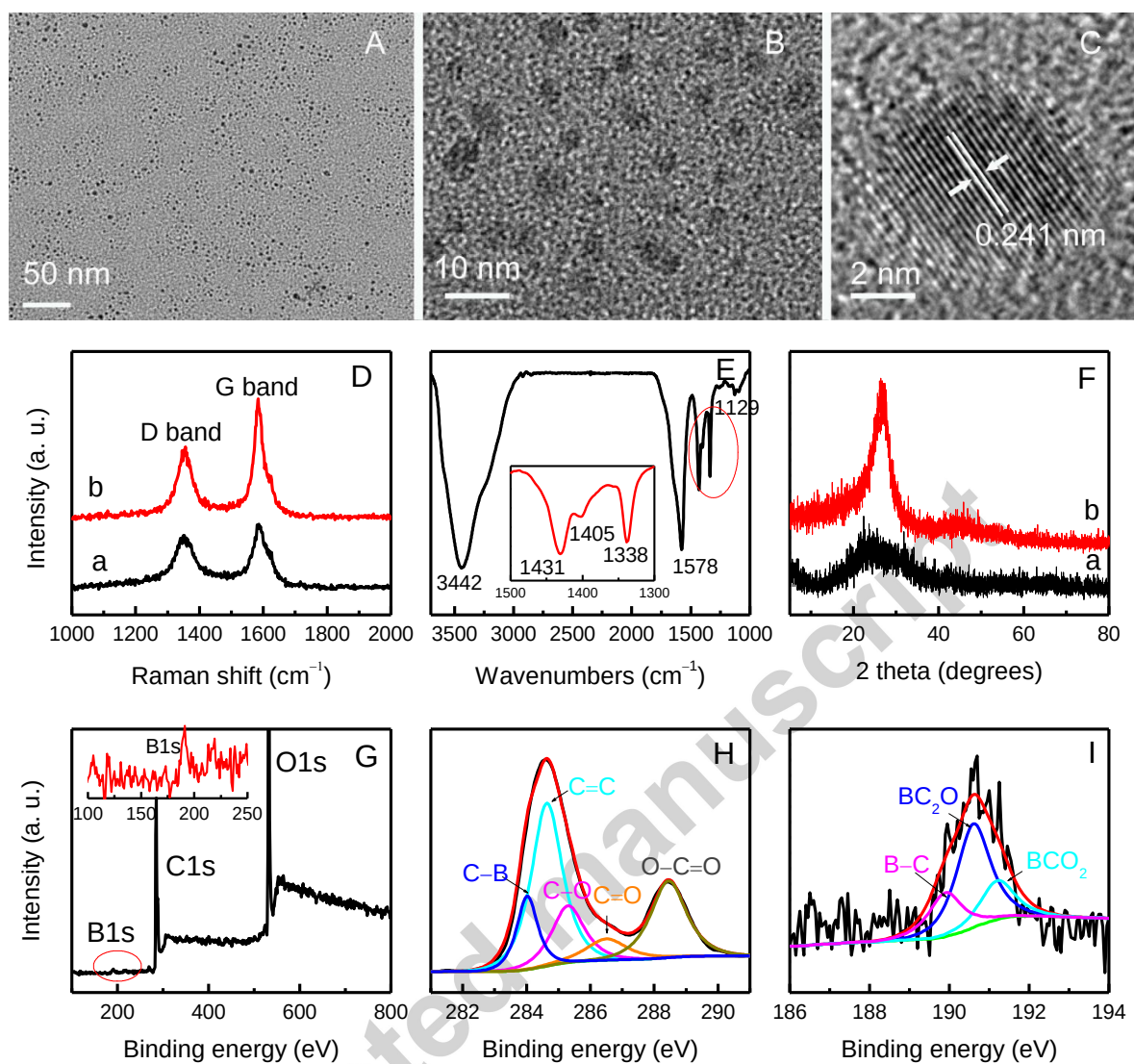


Fig. 1.

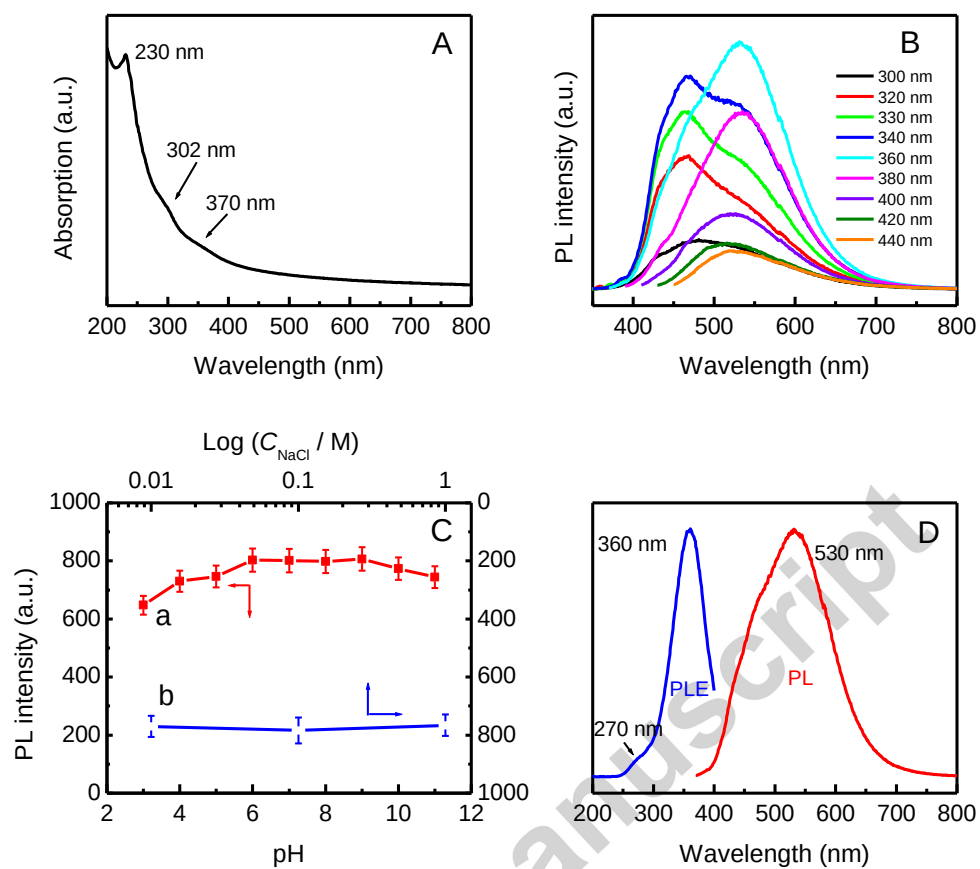


Fig. 2.

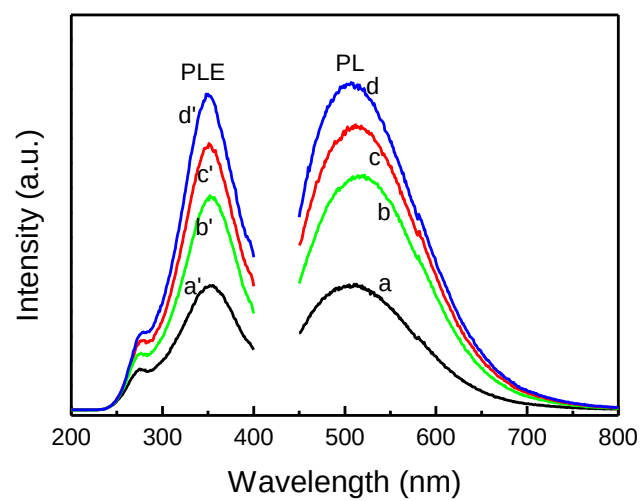


Fig. 3.

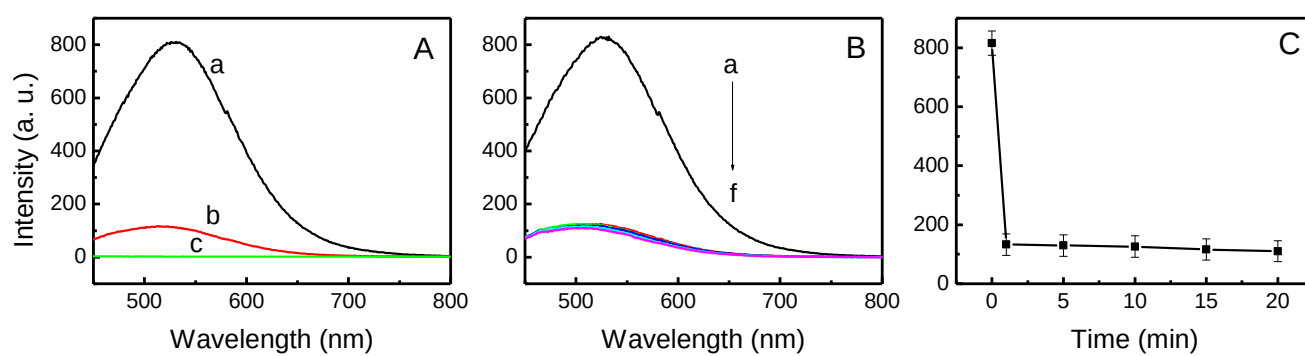


Fig. 4.

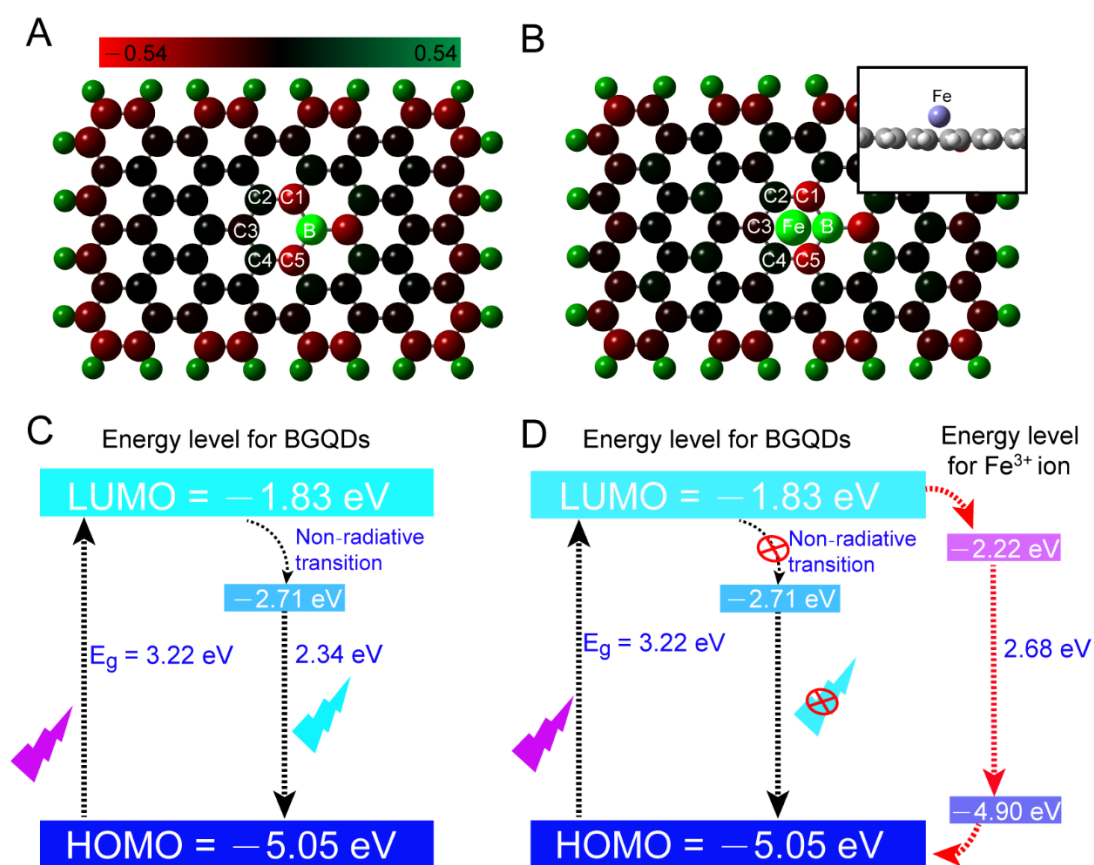


Fig. 5.

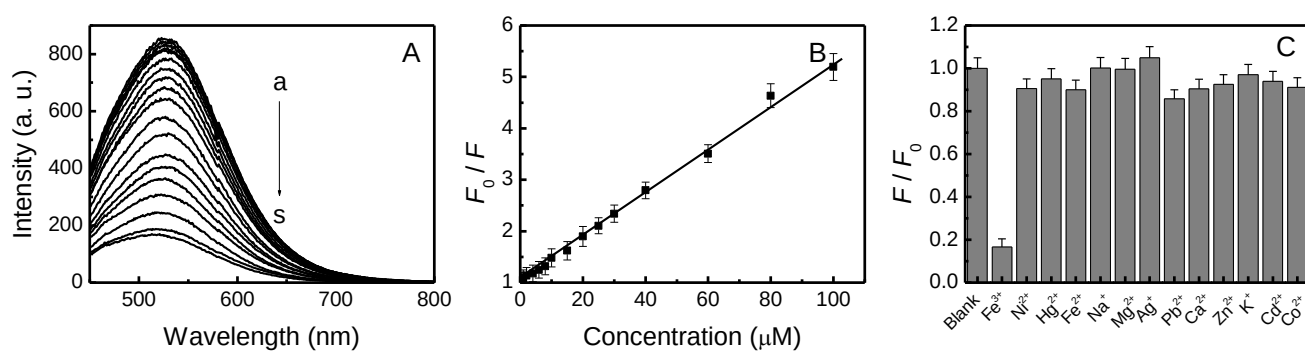


Fig. 6.